

CELEBAL EXCELLENCE INTERNSHIP

WEEK 4 ASSIGNMENT

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CSI_ID - CT_CSI_DE_1099

Mini Project (Final Task)

Problem Statement:

Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated

Summary

The objective of this assignment was to understand the core concepts of Microsoft Azure and build an end-to-end data pipeline using Azure Storage Account and Azure Data Factory (ADF). The implementation covered the complete process of creating cloud resources, configuring storage, connecting services, building a data pipeline, validating metadata, and executing the pipeline successfully.

The assignment began with the creation of a **Resource Group**, which served as a logical container for managing all Azure resources. Next, an **Azure Storage Account** was created, followed by the creation of two Blob Storage containers named **source** and **destination**. A CSV file was uploaded to the source container to act as the input data for the pipeline.

An **Azure Data Factory** instance was then created to orchestrate the data movement process. Inside ADF Studio, a **Linked Service** was configured to establish a secure connection between Azure Data Factory and the Azure Storage Account. Two datasets were created: a **source dataset** pointing to the uploaded CSV file and a **destination dataset** pointing to the destination Blob container.

To validate the input file before processing, a **Get Metadata** activity was added to the pipeline. This activity retrieved important file properties such as file existence, size, and last modified date. After successful metadata validation, a **Copy Data** activity was configured to copy the CSV file from the source container to the destination container. The activities were connected sequentially to ensure that data copying occurred only after successful metadata validation.

The pipeline was validated, published, and executed using the **Debug** option. The execution completed successfully without any errors. The copied CSV file was verified in the destination container, confirming that the pipeline functioned correctly. The pipeline execution status and activity logs were monitored using the **Monitor** section of Azure Data Factory.

Finally, **Azure IAM (Identity and Access Management)** was explored to understand role-based access control. Appropriate permissions such as **Reader** and **Contributor** (or the available assigned role) were reviewed to ensure secure access between Azure Data Factory and Azure Storage.

Result

The Azure Data Factory pipeline executed successfully. The source CSV file was copied from the Blob Storage source container to the destination container after successful metadata validation. This assignment demonstrated the practical implementation of cloud storage, data integration, metadata validation, pipeline orchestration, and monitoring using Microsoft Azure services.

Key Azure Services Used

- Azure Resource Group
- Azure Storage Account
- Azure Blob Storage
- Azure Data Factory (ADF)
- Linked Services
- Datasets
- Get Metadata Activity
- Copy Data Activity
- Azure Identity and Access Management (IAM)

Learning Outcomes

- Understood Azure cloud resource management.
- Learned to create and manage Azure Storage Accounts and Blob Containers.
- Configured Linked Services and Datasets in Azure Data Factory.
- Implemented metadata validation using the Get Metadata activity.
- Built and executed an end-to-end data pipeline using the Copy Data activity.
- Monitored pipeline execution and verified successful data transfer.
- Gained practical experience with Azure IAM and role-based access control.

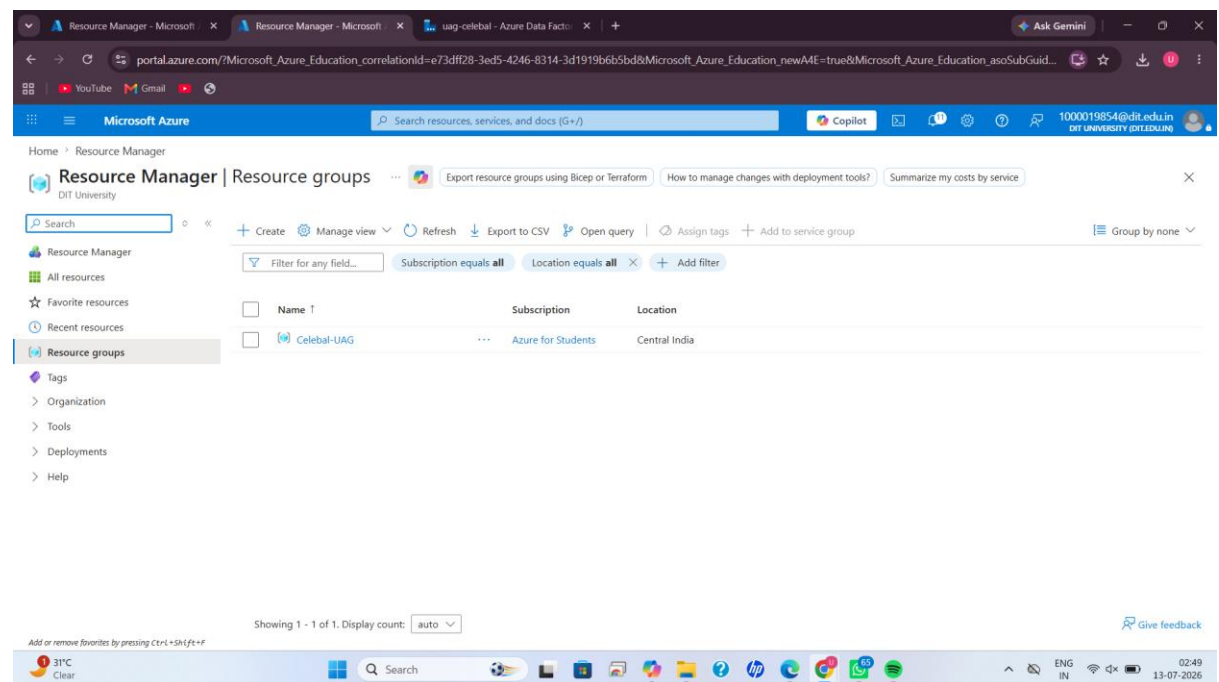
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

Resource Manager - Microsoft | uag-celebal - Microsoft Azure | uag-celebal - Azure Data Factory

adf.azure.com/en/management/datalinkedservices?factory=%2Fsubscriptions%2Fa38f0608-5361-489b-b568-bced9b0957%2FresourceGroups%2FCelebal-UAG%2Fproviders%2FMicrosoft.DataFactory

Microsoft Azure | Data Factory | uag-celebal

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Validate all | Publish all

General

- Connector upgrade advis...
- Factory settings

Connections

- Linked services
- Integration runtimes
- Microsoft Purview
- ADF in Microsoft Fabric

Source control

- Git configuration
- ARM template

Author

- Triggers
- Global parameters
- Data flow libraries

Security

- Credentials

Successfully created

Successfully created AzureBlobStorage1 (Linked service).

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name

Annotations: Any

Showing 1 - 1 of 1 items

Name	Type	Related	Annotations
AzureBlobStorage1	Azure Blob Storage	0	

Resource Manager - Microsoft | unnatstorage123 - Microsoft Azure | uag-celebal - Azure Data Factory

portal.azure.com/?Microsoft_Azure_Education_correlationId=e73dff28-3ed5-4246-8314-3d1919b6b5bd&Microsoft_Azure_Education_newA4E=true&Microsoft_Azure_Education_asoSubGuid=...

Microsoft Azure

Home > Storage center | Blob Storage > unnatstorage123

Storage center | Blob Storage

DIT University (dit.edu.in)

Search

Summary | Resources

Overview

All storage resources

- Object storage
- File storage
- Block storage
- Data management
- Migration
- Partner solutions
- Management services
- Help

Create

Name

unnatstorage123

unnatstorage123 | Containers

Storage account

Search

Add container | Upload | Refresh | Delete | Change access level

Search containers by prefix

Only show active containers

Showing all 3 items

Name	Last modified	Anonymous access level	Lease state
slogs	13/07/2026, 01:51:05	Private	Available
destination	13/07/2026, 02:07:53	Private	Available
source	13/07/2026, 02:07:38	Private	Available

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

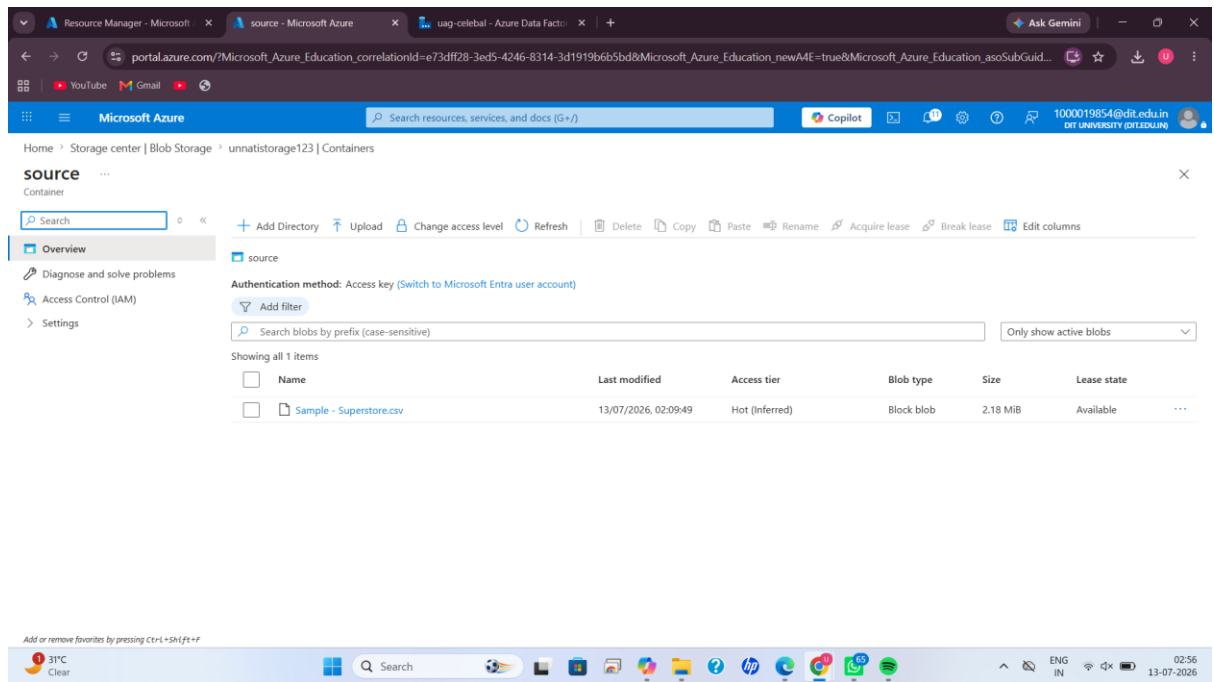
Queues

Tables

Security + networking

Showing 1 - 1 of 1. Display count: auto

Add or remove favorites by pressing Ctrl+Shift+F



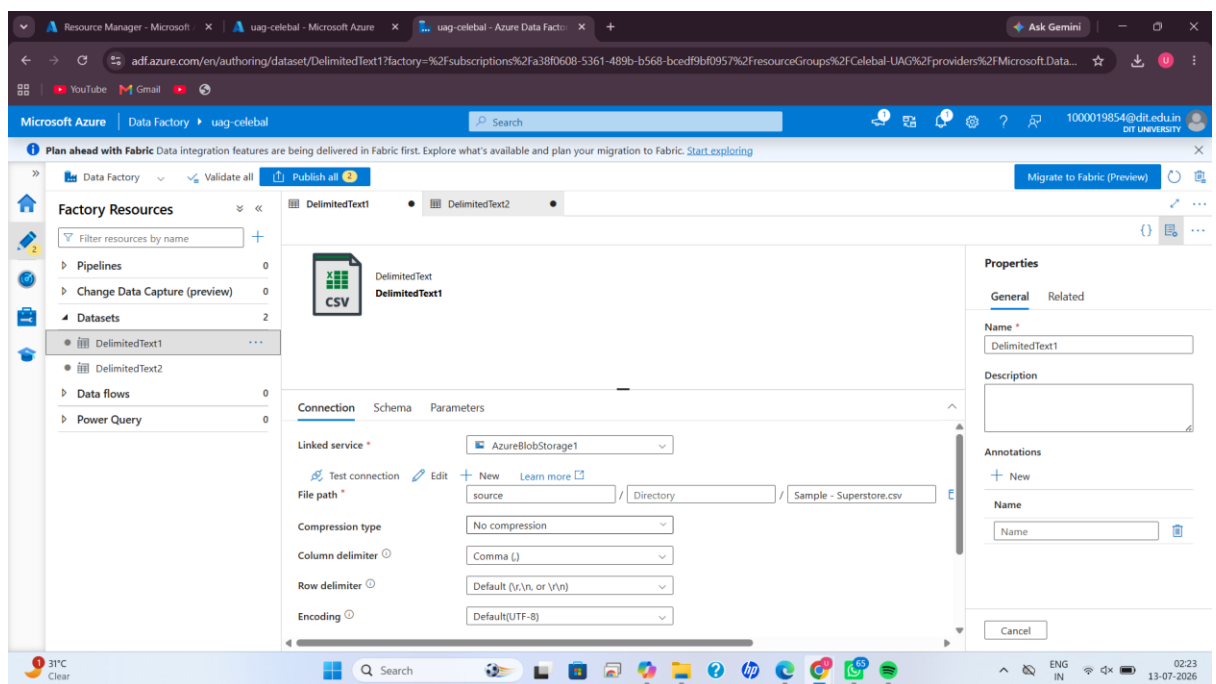
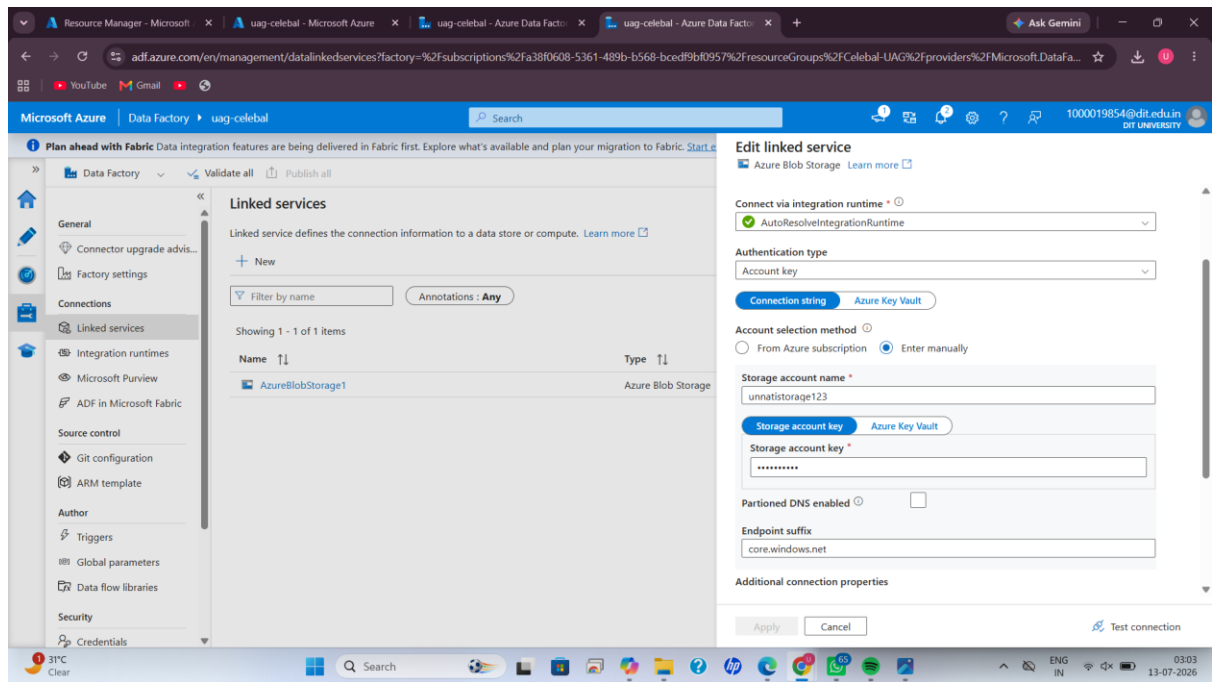
Task 3: ADF Basics

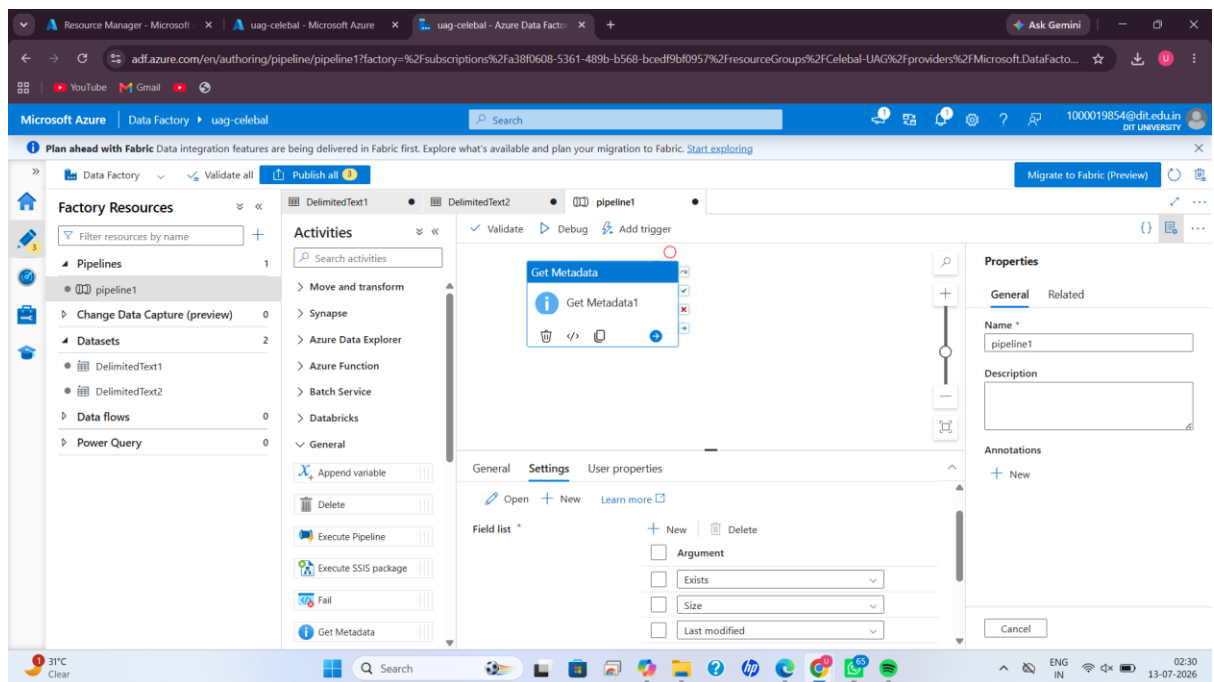
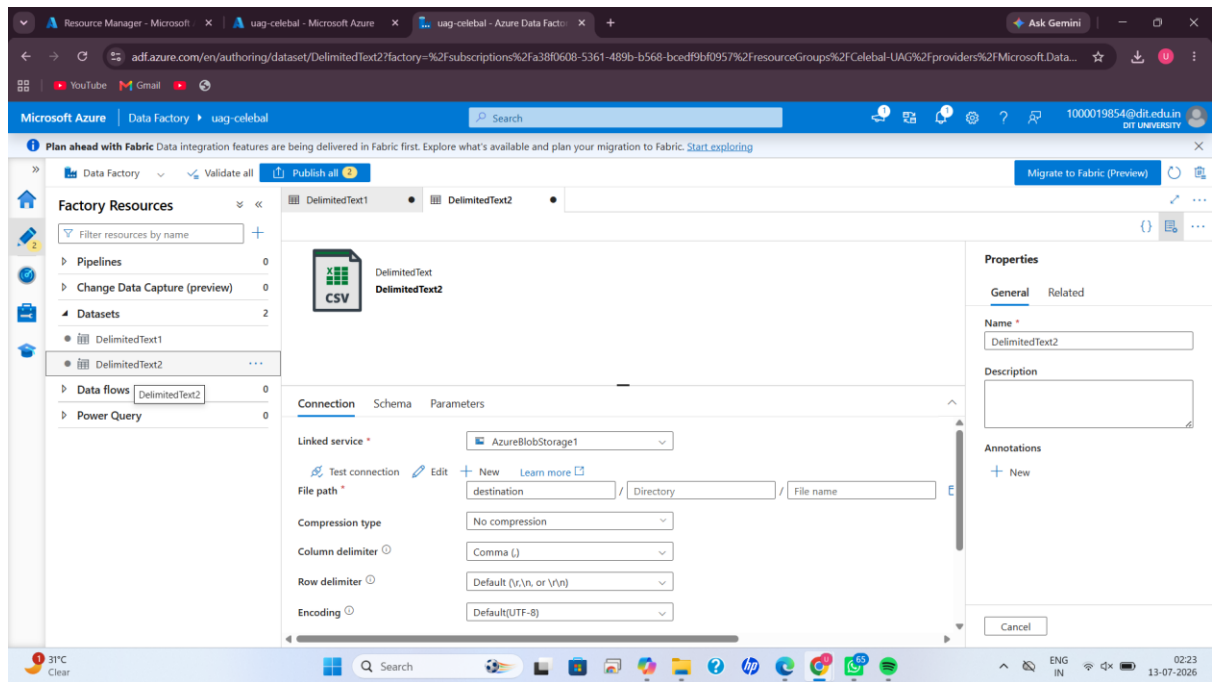
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity





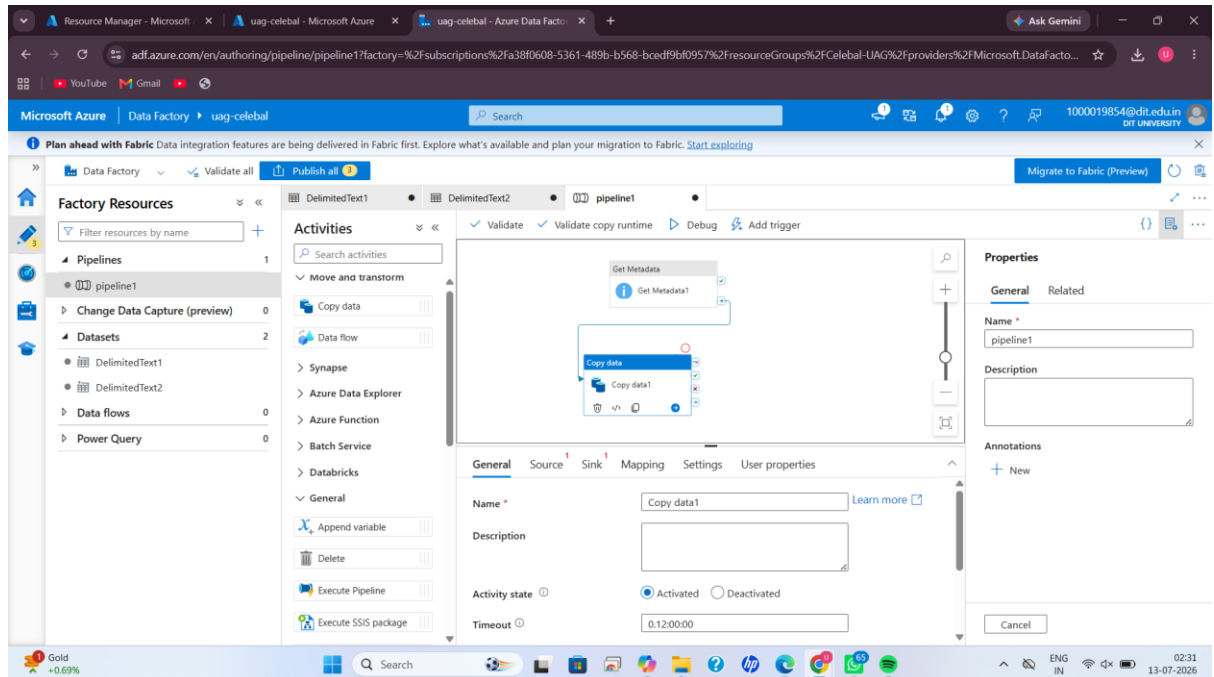
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



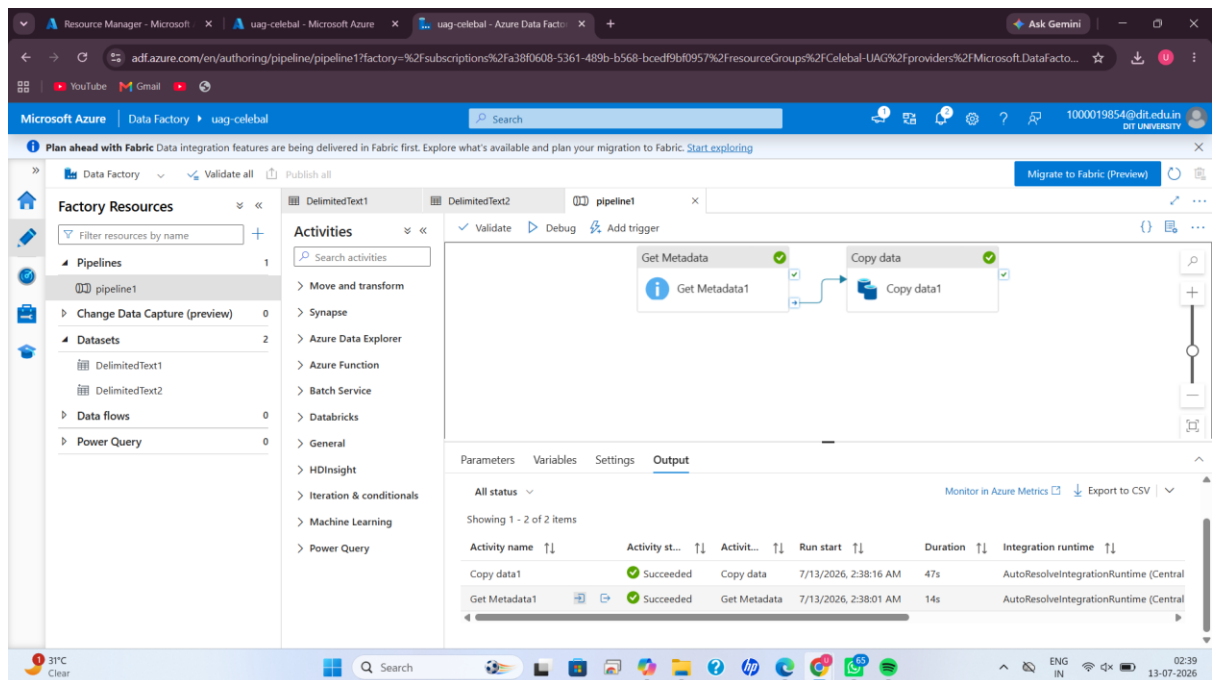
Task 5: Pipeline Execution

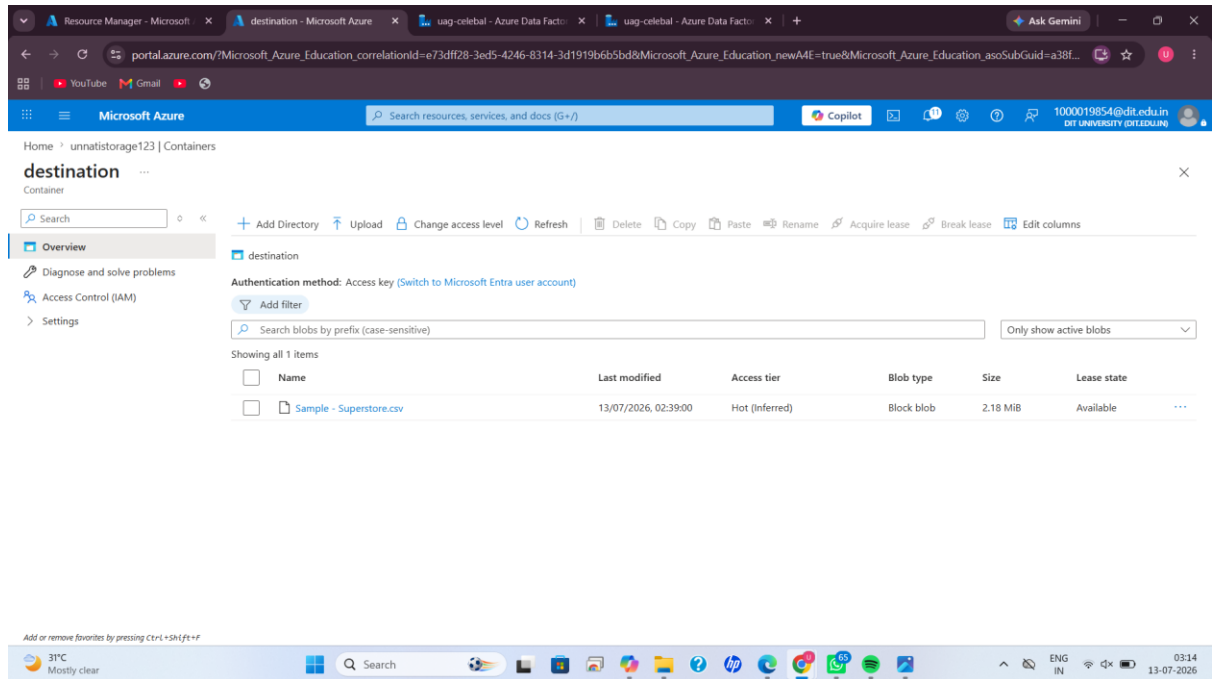
Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)





Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

Resource Manager - Microsoft

unnatistorage123 - Microsoft

uag-celebal - Azure Data Factor

+

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Microsoft Azure

Search resources, services, and docs (G+)

Copilot

1000019854@dit.edu.in
DIT UNIVERSITY (DITEDU.IN)

Homeunnatistorage123

unnatistorage123 | Access Control (IAM)

Storage account

Search

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

Add or remove favorites by pressing Ctrl+Shift+F

+ Add

Download role assignments

Edit columns

Refresh

Delete

Feedback

Check access

Role assignments

Roles

Deny assignments

Number of role assignments for this subscription

24000

Search by name or email

Type: All

Role: All

Scope: All scopes

Group by: Role

All (2)

Job function roles (0)

Privileged administrator roles (2)

☐

Name

Type

Role

Scope

Condition

Owner (2)

☐

Unnati Agarwal
1000019854@dit.edu.in

User

Owner

Subscription (Inherited)

None

☐

Unnati Agarwal
1000019854@dit.edu.in

User

Owner

Subscription (Inherited)

None

Showing 1 - 2 of 2 results.

31°C
Clear

Search

ENG
IN

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13-07-2026